

The Wavepool Hypothesis: A Non-Explosive Relational Cosmology Arising from Three Intangible Fundamentals (V 2)

*Version 2 – Substantially Revised with Refined Mathematical Formulation,
Tighter Conceptual Structure, and Stronger Links to Observational
Cosmology*

*D. Tranberg
Independent Researcher*

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Introduction

A growing body of anomalous observations now challenges several core assumptions of the standard hot Big Bang model. The Wavepool Hypothesis offers a minimalist alternative: the entire observable universe emerges directly from the interplay of just three intangible fundamentals — Mathematics as pure relational structure, Vibration as oscillatory dynamics, and Entropy as intrinsic directional irreversibility — without requiring an explosive hot Big Bang.

Recent data from the James Webb Space Telescope have revealed surprisingly luminous red dwarfs and massive early galaxies at redshifts corresponding to only 300–700 million years after the Big Bang — structures that appear too mature for a uniformly hot early universe. Combined with the persistent Hubble tension and hints of evolving dark energy, these findings suggest the early cosmos may have been significantly patchier and less thermally uniform than previously thought.

The Wavepool Hypothesis therefore presents a coherent alternative: a cooler, non-explosive emergence driven solely by the three fundamentals that can reproduce many of the observable features of the standard Big Bang while differing profoundly in its underlying mechanism and early-time dynamics (see the minimal dynamical toy model in Part 3).

Section 1: Pre-Geometric Potential Space – The Three Fundamentals

The question of how an unformed potential becomes an actual, observable universe is one of the oldest and deepest problems in philosophy and physics. Humanity has long recognized that we can conceive of far more possibilities than are ever expressed in physical reality. Thinkers from Aristotle to Wheeler, Tegmark, and proponents of ontic structural realism have each wrestled with the transition from a pre-geometric substrate to the structured world we inhabit.

The Wavepool Hypothesis addresses this question in a minimalist way. Prior to the Emergence Event, reality consists of a purely relational substrate — Potential Space — that cannot be described by any classical mathematical structure such as a manifold, Hilbert space, or set-theoretic object. This substrate is governed solely by three irreducible, co-equal fundamentals in their unrealized form:

- **Mathematics** — *pure relational structure, the vertical lattice that defines all possible configurations and entanglement relations.*
- **Vibration** — *oscillatory dynamics, present as unrealized high/low bi-pole potential.*
- **Entropy** — *intrinsic directional irreversibility, present as unrealized directional potential that will later manifest both as the internal arrow of time and as the classical expansive expression of dark energy(realized dynamically as the primitive bias parameter δ in the toy model of Pt 3).*

In this pre-geometric realm the three fundamentals exist as an intimately intertwined relational substrate. Mathematics provides the governing framework within which Entropy and Vibration form a complementary pair. No external energy or additional mechanism is required. The entire system remains in a state of undifferentiated potential until the relational structure of Mathematics enables Entropy's irreversible differentiation to act upon Vibration, releasing the accumulated relational tension inherent in the three-fundamental system. The precise dynamical realization of this transition is given by the coupled energy-cycle equations in Part 3, Section 3.

Section 2: The Wavepool Concept

The name “Wavepool Hypothesis” is more than a convenient metaphor; it is the central organizing concept of the model. Just as a water park wave pool uses hidden pumps to build immense potential energy beneath a calm surface — with vibrational modes held in check until the gate is opened — the pre-emergent relational substrate uses the three fundamentals to hold vibrational modes in perfect

destructive interference within mathematical constraints. Entropy stands ready as the directional force that will break this cancellation. When the Emergence Event occurs, the stored relational tension is released, producing interfering waves that amplify into localized high-amplitude structures while dissipating energy through expansion — direct analogs of early nucleosynthesis and structure formation in the Turbulent Foam Dynamics regime. This single medium — the primal quantum foam — generates all subsequent complexity, from the instantiation of spacetime to cosmic expansion, without requiring additional mechanisms or a global high-temperature phase.

Section 3: The First Trigger – Spontaneous Emergence

At the Emergence Event, the intrinsic relational nature of Mathematics acts upon both Entropy and Vibration. Until that moment, Entropy's directional irreversibility had remained an unrealized state, held within the vertical principle of Mathematics together with the vibrational dualistic modes, which were locked in perfect destructive interference.

This transition from a pre-geometric substrate to classical reality echoes foundational ideas in quantum cosmology and pregeometry (*Wheeler 1960s–70s; Hartle & Hawking 1983; Vilenkin 1982*), though Wavepool approaches the trigger through a purely relational triad rather than quantum-gravitational or Euclidean path-integral methods. At the moment of action, the constraint is lifted and the stored relational tension is released, instantly actualizing a classical pocket universe. Vibration, now free to respond to Entropy's directional bias, immediately co-instantiates throughout the new pocket universe. As it does so, it forms eddies and tides whose localized concentrations eventually solidify into high-pole classical mass and material structures. The vast majority of the vibration, however, spreads out as a naturally occurring low-vibrational pole background environment — a diffuse, universe-wide relational background that continues to exist within the continually expanding spacetime fields.

From this moment onward, a dynamic we are labeling the *universal Vibrational pump* — the slow, continuous relational transfer between the low-vibration background energy *EL* and *high-vibration localized energy EH*, governed by the entropy bias $\delta > 0$ — becomes active (see explicit coupled ODEs in Part 3, Section 3). A tiny fraction of the low-vibrational background is steadily promoted into high-vibrational localized matter, while dissipation processes (radiation, decay, redshift) return energy to the background reservoir. Entropy enforces the overall direction of the cycle, ensuring the process is irreversible and self-sustaining. No external energy input is ever required; the entire system is closed and conserves total relational energy. Cosmic expansion itself is driven directly by the low-vibration background ($H = \alpha EL$).

Section 4: Early Classical Structure

Once the pocket universe has been instantiated through the entropic wavefront, the three fundamentals manifest as a unified scalar field Φ_{total} within the new classical reality. This scalar field is the simplest classical structure capable of expressing their relational interplay: Mathematics supplies the relational lattice, Vibration supplies the oscillatory energy and dynamics, and Entropy supplies the directional bias. The fundamentals themselves remain three distinct primitives.

The field adopts a *Mexican-hat-like effective potential* (whose dynamical realization appears as the instability in the toy-model field equation of Part 3):

$$V(\Phi_{total}) = -2K\Phi_{total}^2 + 4\lambda\Phi_{total}^4 + \epsilon Entropy\Phi_{total}.$$

The entropic term tilts the potential, ensuring the low-vibrational pole background environment — the vast cosmic ocean of diffuse vibration — initially dominates and drives rapid expansion. As spacetime expands and density dilutes, the high-vibrational pole gains relative strength. Localized resonances begin to form, producing the first stable high-vibrational excitations that will eventually become the primitive particles of the emerging universe.

The system evolves according to the relativistic Klein-Gordon equation (the continuum limit of the toy-model field dynamics in Part 3):

$$\ddot{\Phi}_{total} + 3H\dot{\Phi}_{total} + d\Phi_{total}/dV = 0.$$

The universal Vibrational pump now operates continuously through the pervasive quantum vacuum fluctuations and the diffuse classical baryonic ocean. This interplay conserves total relational energy while allowing the low-vibrational pole background environment to dominate as the expansive dark-energy component and the high-vibrational pole to gradually localize into gravity-bound classical matter. Entropy's intrinsic irreversibility ensures the overall flow remains one-way on cosmic timescales.

Gravity itself emerges directly from the three fundamentals in the earliest instants after the Emergence Event. The immense energy density released in the transition drives an immediate distortion of the relational structure governed by Mathematics. This distortion, interacting with Entropy's directional bias, produces the first gradients in the nascent geometry. Gravitational waves — the earliest dynamical excitations of the emergent spacetime — begin to stir and differentiate the medium even before stable high-vibrational structures form. These constitute the first waves of reality itself and propagate eternally throughout the expanding universe as part of the low-vibrational pole background environment. As high-vibrational modes eventually localize into classical matter (*via localized peaks in the toy-model field $\phi(x,t)$*), they create regions of significantly higher relational density within the otherwise diffuse low-vibrational pole background environment. The resulting curvature of spacetime is gravity — the natural geometric response to the presence of high-vibrational energy.

Section 5: Turbulent Foam Dynamics (TFD) Regime and Structure Formation

A quantitative description of the pump-driven matter production and gravitational collapse processes described below will be developed in Wavepool 4. In the wake of the Emergence Event, the immense energy density released into the nascent spacetime drives violent primordial gravitational waves — the first dynamical excitations of the emergent geometry. These waves vigorously stir the medium, creating the Turbulent Foam Dynamics regime that persists for approximately the first 380,000 years. Within this dense, convecting foam, constructive interference produces transient regions of extremely high field strength. These localized resonances (*emergent from the ϕ -field instability in the Part-3 toy model*) act as miniature versions of the critical field required for Schwinger-like pair production, converting virtual particle–antiparticle pairs from the low-vibrational pole background environment into real particles and baryons.

Early in this manifestation phase the extreme high/low pressure waves dominate the initial production of free-floating particles and proto-elemental material. As the initial emergence slows and the medium expands and cools, the universal Vibrational pump becomes the dominant mechanism of matter production. The efficiency of this process is low — only a tiny fraction of the low-vibrational pole background environment is converted into stable matter — but the creation is inherently clumpy rather than uniform. This clumpiness arises naturally from the wave interference and turbulent currents within the foam (*directly reproduced by localized peaks in the toy-model vibrational field $\phi(x,t)$*).

Within these localized high-vibrational resonances, the same entropic directional bias that produces the global dominance of the low-vibrational pole background environment also introduces a small but systematic preference for matter over antimatter during Schwinger-like pair production. This bias arises directly from Entropy’s intrinsic directional irreversibility acting upon the vibrational modes, providing the out-of-equilibrium and effective CP-violating conditions required for baryogenesis.

As the medium continues to expand and cool, quarks and leptons progressively bind into stable protons, neutrons, and eventually atoms. By the time the temperature reaches approximately 3000 K (recombination), a temporary outer shell of cooler, denser material has formed around the expanding pocket universe. Gravitational instabilities then cause this shell to fracture gradually along tension lines between gravitational compression and the dark-energy-driven expansion supplied by the low-vibrational pole background environment. The primal dense ball slowly differentiates and fragments in a patchy manner. Some fragments collapse under local gravity into the first stars, black holes, and planetary seeds, while the remaining detritus expands outward as a diffuse cloud of gas and dust.

This patchy differentiation naturally accounts for the surprisingly early appearance of low-metallicity luminous red dwarfs (LRDs), overmassive early black holes, and other anomalous structures observed by JWST, without requiring a global high-temperature phase.

Section 6: Testable Predictions

The Wavepool Hypothesis makes several distinctive, near-term testable predictions that arise directly from its core mechanisms: the entropic wavefront at the Emergence Event, the strong bias toward the low-vibrational pole background environment, the universal Vibrational pump operating as a continuous cycle from the earliest instants (as formalized in the Part-3 toy model), and the patchy, localized high-field resonances sustained by primordial gravitational waves in the Turbulent Foam Dynamics regime.

- Primordial gravitational wave background at *ultra-low frequencies* ($\sim 10^{-16}$ to 10^{-15} Hz). The immense energy density released at the Emergence Event generates violent primordial gravitational waves — the first dynamical excitations of the emergent spacetime. These waves, together with the subsequent stirring in the TFD regime, should produce a stochastic gravitational-wave background at extremely low frequencies. Even non-detections or upper limits from LISA in the 2030s would meaningfully constrain the energy scale and timing of the Emergence Event.
- Patchy CMB spectral distortions (*μ -type and y -type*). Localized energy injection and dissipation in high-field resonances during the TFD regime should produce spatially inhomogeneous distortions with amplitudes in the range 10^{-8} to 10^{-6} , significantly patchier than those expected from a uniform hot Big Bang. These should be detectable by the Simons Observatory (through 2027) and measurable in detail by CMB-S4.
- Primordial non-Gaussianities and early black-hole seeds. The localized high-vibrational resonances and gravitational instabilities driven by the universal Vibrational pump and primordial gravitational waves naturally produce *moderate positive non-Gaussianity* ($f_{NL} \sim +5$ to $+30$) and an *enhanced population* of direct-collapse black hole candidates at high redshift. These signatures should be observable in detailed JWST and Roman Space Telescope surveys of early structure.
- Time-varying dark energy. The low-vibrational pole background environment is predicted to slowly relax through the universal Vibrational pump, producing a mild evolution in the dark-energy equation-of-state parameter of the form $w = -1 + \epsilon(z)$ with $|\epsilon| \sim 0.01\text{--}0.05$ at low redshift. This evolution should be detectable by future Euclid supernova and galaxy-clustering data, extending the mild hints already seen by DESI.

- Topological or parity-violating signatures in the *CMB*. The advancing entropic wavefront imposes a global topological differentiation on the vibrational field that may leave faint parity-violating patterns or hemispherical asymmetries in *CMB* polarization at the level of 10^{-6} to 10^{-5} . Future polarization missions (*CMB-S4* and *LiteBIRD*) are expected to reach the sensitivity required to detect or rule out such anomalies.

These predictions differ from standard Λ CDM expectations in ways that are directly traceable to the three-fundamental mechanisms and the toy-model dynamics.

Section 7: Conclusion and Outlook

The Wavepool Hypothesis proposes that the observable universe emerges from a pre-geometric relational substrate governed by three intangible fundamentals: Mathematics as pure relational structure, Vibration as oscillatory dynamics, and Entropy as intrinsic directional irreversibility. At the Emergence Event, these fundamentals interact to produce a relatively cool, patchy actualization of spacetime, matter, and the arrow of time. This is followed by a prolonged Turbulent Foam Dynamics regime in which the universal Vibrational pump operates continuously between the low-vibrational pole background environment and the localized high-vibrational pole (as captured by the minimal dynamical toy model of Part 3), giving rise to the patchy differentiation and structure formation we observe today.

This interpretation builds directly upon the rich body of observational data painstakingly gathered by generations of researchers who worked within the standard hot Big Bang framework. We are deeply grateful for that foundational work. The cosmic microwave background measurements, nucleosynthesis constraints, large-scale structure surveys, and more recent JWST and DESI observations represent an extraordinary empirical achievement. The present model does not reject this data; rather, it offers an alternative interpretive framework that seeks to explain the same observations through a non-explosive, patchy emergence driven by the three fundamentals.

Some features that have proven difficult to reconcile with a uniform hot Big Bang — such as the surprisingly early appearance of massive, low-metallicity luminous red dwarfs, overmassive black holes at high redshift, hints of evolving dark energy, and the persistent Hubble tension — appear more naturally accommodated within the Wavepool picture of localized high-vibrational pockets and continuous cyclic differentiation driven by the low-vibrational pole background environment. In this sense, Wavepool does not discard earlier data; it suggests a different way of understanding what that data has been telling us.

The hypothesis remains exploratory and is offered with humility. It is falsifiable through specific, near-term observations, including patchy CMB spectral distortions, ultra-low-frequency primordial gravitational waves from the Emergence Event itself, primordial non-Gaussianities, and continued evolution of the dark-energy equation-of-state parameter. The general critiques and open challenges outlined in Section 6 — particularly the need for quantitative derivations of the universal Vibrational

pump rate, the vacuum-energy suppression, and the precise origin of the baryon asymmetry — highlight the substantial work that remains. These issues will be the central focus of the planned Wavepool 4 paper.

We invite the scientific community to examine, critique, and test this framework. Whether the Wavepool Hypothesis ultimately stands or falls, we hope the ideas presented here contribute constructively to the ongoing effort to understand our cosmic origins. The pursuit of truth in cosmology benefits from diverse perspectives, and we are grateful for the opportunity to offer one more voice in that shared endeavor.

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